KPC Inuka Fellowship Hackathon 3

1. Prologue: KPC – PLC

Kenya Pipeline Company PLC (KPC PLC) is a publicly listed company, following its listing on the Nairobi Securities Exchange.

Established in 1973 under the Companies Act (**Cap 486 of the Laws of Kenya**), the Company commenced commercial operations in February 1978. Over the years, KPC PLC has evolved into a critical national infrastructure utility supporting energy security and regional economic integration.

KPC PLC is mandated to provide safe, reliable, efficient, and cost-effective transportation, storage, and distribution of petroleum products from the Port of Mombasa to inland markets in Kenya and the broader East African region. To deliver on this mandate, the Company has developed and continues to maintain an extensive infrastructure network comprising approximately 1,342 kilometres of pipeline, along with strategically located storage, pumping, and loading facilities. This system has a combined capacity of approximately 14 billion litres of petroleum products annually.

As a listed entity, KPC PLC is committed to high standards of corporate governance, transparency, and shareholder value creation, while maintaining its public service mandate in support of national development objectives.

The Company's strategic direction is guided by the KPC Vision 2030 Strategic Plan, which is anchored on four strategic pillars:

- **Diversification and Growth** – Driving long-term sustainability and expansion, including positioning the Company for resilience and opportunity within the global transition to cleaner and more sustainable energy systems.

- **Operational Excellence** – Enhancing efficiency, reliability, and competitiveness through continuous improvement, digitization, and adoption of advanced technologies.
- **Environmental, Social and Governance (ESG)** – Embedding ESG principles across operations to ensure responsible stewardship, compliance, sustainability, and long-term stakeholder value creation.
- **People** – Strengthening human capital through talent development, capability building, employee engagement, and productivity enhancement to support strategic delivery.

2. Problem Statement for Hackathon 3

Title: The Autonomous Supply Chain & Executive Control Plane

Date: Wednesday, 9th-11th September 2026

Theme: *"Ship, Secure, and Scale: From Predictive Insights to Production-Ready Automation."*

Objective: Teams will transition their Stage 2 analytical prototypes into hardened, deployable, AI-powered production services. The goal is to prove that the solution can survive the rigors of KPC's live downstream operations, integrate seamlessly with existing workflows, and deliver a quantifiable Return on Investment (ROI) to executive leadership.

Deliverables for Stage 3:

- **Production CI/CD & Monitoring**: The solution must be deployed via a mature CI/CD pipeline to a live, monitored environment with automated alerting (e.g., PagerDuty/Slack integrations for system failures or threshold breaches).
- T**he Executive Control Plane**: A polished, executive-facing dashboard that translates complex operational data into strategic business metrics (e.g., "KES Saved in Demurrage," "Throughput % Increase").
- **Handover & Support Readiness**: Comprehensive documentation, including an operations manual, disaster recovery plan, and a formal support handover plan for the Em-Tech/KPC IT teams.

- **The ROI Business Case**: A rigorously quantified financial narrative detailing the implementation costs versus the operational savings (e.g., reduced truck turnaround time, prevented stock-outs).
- **Presentation**: A 10-minute live demo and ROI pitch + extended Q&A, pitched directly to KPC and Em-Tech executive leadership.

3. PROBLEM DOMAINS

Core Challenges Clustered into 5 Domains (10 Problem Statements)

The core challenges revolve around KPC's Downstream Supply Chain, Depot Operations, and Refinery Integration. To fit the hackathon structure, these have been extrapolated into five specialized domains, each containing two distinct problem statements (10 total). Teams will select one domain and solve one problem within it.

Domain 1: Depot Loading & Yard Operations Optimization

***Focus**: Maximizing physical throughput and reducing bottlenecks at the loading gantry.*

1. **Loading Gantry Cycle-Time Reduction**: Develop an analytics-driven scheduling and sequencing tool to optimize crane/pump loading cycles, minimizing truck turnaround time and idle periods at the gantry.
2. **Yard Traffic & Queuing Intelligence**: Create a real-time yard management system that predicts bottlenecks, dynamically routes trucks within the depot, and reduces physical queuing costs and congestion.

Domain 2: Automated Dispatch & Fleet Scheduling

***Focus**: Transitioning from manual scheduling to intelligent, automated fleet orchestration.*

3. **AI-Driven Fuel Batch Truck Scheduling**: Build an automated scheduling engine that matches fuel batch availability from the refinery with truck capacity and delivery windows, eliminating manual spreadsheet-based dispatching.

4. **Dynamic Route & Demurrage Prevention**: Develop a predictive routing and time-window optimization tool that accounts for traffic, depot congestion, and customer receiving hours to reduce demurrage (delay) penalties drastically.

Domain 3: Downstream Refinery & Pipeline Integration

***Focus**: Synchronizing the physical flow of product from the refinery output to the regional depots.*

5**. Refinery-to-Depot Flow Synchronization**: Create a digital control model that balances refinery output rates with depot tank farm capacities, preventing both refinery throttling (due to full depot tanks) and depot stock-outs.

6. **Downstream Demand Forecasting & Inventory Replenishment**: Build a predictive inventory model that anticipates regional fuel demand spikes and automatically triggers pipeline batching or truck dispatch to maintain optimal stock levels.

Domain 4: Supply Chain Revenue & Contract Compliance (Fintech/Operations)

***Focus**: Ensuring that operational efficiencies translate directly into protected revenue and contract compliance.*

7. **Automated Demurrage & Detention Billing Engine**: Develop a system that automatically calculates, validates, and invoices demurrage charges based on the truck scheduling and yard tracking data, closing revenue leakage.

8. **Loading Manifest vs. Invoiced Volume Reconciliation**: Build an automated reconciliation pipeline that compares physical loading meter data against dispatched invoices and payments, flagging discrepancies in real-time.

Domain 5: HSE & Asset Reliability in Depot/Refinery Operations

***Focus**: Ensuring that increased throughput and automation do not compromise safety or equipment lifespan.*

9. **Loading Arm & Pump Predictive Maintenance**: Utilize telemetry data to anticipate failures in high-wear depot equipment (loading arms, pumps, valves), shifting to condition-based maintenance to prevent unplanned downtime.

10. **Real-Time Spill & Overfill Prevention Analytics**: Develop an early-warning HSE system that monitors tank levels and flow rates during loading, automatically triggering alerts or shutdowns to prevent environmental spills and overfills.

4. The "Linkup": Upgrading Hackathon 2 for Stage 3

The Concept: "From Predictive Insights to the Autonomous Control Plane"

- **Hackathon 1 (Data)**: Built the pipes to move messy data.
- **Hackathon 2 (Analytics)**: Built the brain (dashboards, predictive models for truck scheduling, throughput forecasting).
- **The Stage 3 Linkup/Upgrade**: The upgrade transitions the Stage 2 "**Brain**" into an "**Autonomous Control Plane**."
    - ***The Shift***: In **Stage 2**, a predictive model might *show* a dashboard alerting managers that a truck will be delayed. In the **Stage 3 Linkup**, that model must be upgraded to *act*, automatically triggering an API that reschedules the truck, reroutes yard traffic, or adjusts the refinery batch flow without human intervention.
    - ***The Focus***: The upgrade is not about writing new models; it is about wrapping **Stage 2** models in production-grade APIs, event-driven architectures, and automated feedback loops, turning passive analytics into active, resilient operational automation.

5. Grading Rubric for Hackathon 3

This rubric reflects the Stage 3 weighting (45% of the total series score) and strictly evaluates the "Platform, Quality & Deployment" layer as defined in the roadmap.

Criterion	Weight	What Judges Look For (Stage 3 Focus)
Deployment, Quality & Resilience	30%	The Hardening Check: Is the solution deployed via a robust CI/CD pipeline? Are there automated alerts, error-handling, and monitoring in place? Is the system resilient enough for 24/7 KPC downstream operations?
Quantified ROI & Business Case	25%	The Executive Value: Is the financial impact rigorously quantified (e.g., KES saved, hours recovered)? Are the assumptions honest and backed by the data? Does the narrative prove the solution pays for itself?
Integration & Usability	20%	The Workflow Fit: Does the tool integrate smoothly with KPC's existing systems (via APIs)? Is the UX intuitive for both the depot floor operators and the C-suite executives viewing the control plane?
Documentation & Handover	15%	The Sustainability Check: Could an Em-Tech or KPC engineer take over this system tomorrow? Are the operations manual, support plan, and codebase documentation complete and professional?
Pitch & Executive Storytelling	10%	The Leadership Communication: Is the 10-minute pitch confident, concise, and focused on business value rather than just technical features? Does the team handle executive Q&A with operational maturity?

Judging Panel Composition for Stage 3:

- **KPC Executive Leadership / Board Representatives**: To evaluate ROI, strategic alignment, and adoption readiness.
- **Em-Tech CTO / Senior Engineering Leads**: To evaluate CI/CD maturity, system resilience, and code quality.
- **PLP Academy Principal**: To evaluate the fellows' growth, presentation skills, and transition from "**student**" to "**industry professional**."